

Questions

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Q1 - 2024 (04 Apr Shift 1)

Let $f(x) = \begin{cases} -2, & -2 \leq x \leq 0 \\ x-2, & 0 < x \leq 2 \end{cases}$ and $h(x) = f(|x|) + |f(x)|$. Then $\int_{-2}^2 h(x) dx$ is equal to :

- (1) 1
- (2) 6
- (3) 4
- (4) 2

Q2 - 2024 (04 Apr Shift 1)

If $\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{1 + \sin x \cos x} dx = \frac{1}{a} \log_e \left(\frac{a}{3} \right) + \frac{\pi}{b\sqrt{3}}$, where $a, b \in \mathbb{N}$, then $a + b$ is equal to _____

Q3 - 2024 (04 Apr Shift 2)

If the value of the integral $\int_{-1}^1 \frac{\cos \alpha x}{1+3^x} dx$ is $\frac{2}{\pi}$. Then, a value of α is

- (1) $\frac{\pi}{3}$
- (2) $\frac{\pi}{6}$
- (3) $\frac{\pi}{4}$
- (4) $\frac{\pi}{2}$

Q4 - 2024 (05 Apr Shift 1)

The value of $\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y} dy$ is :

- (1) $2\pi^2$
- (2) $\frac{\pi^2}{2}$
- (3) $\frac{\pi}{2}$
- (4) π^2

Q5 - 2024 (05 Apr Shift 1)

The integral $\int_0^{\pi/4} \frac{136 \sin x}{3 \sin x + 5 \cos x} dx$ is equal to :

Questions

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(1) $3\pi - 50 \log_e 2 + 20 \log_e 5$

(2) $3\pi - 25 \log_e 2 + 10 \log_e 5$

(3) $3\pi - 10 \log_e (2\sqrt{2}) + 10 \log_e 5$

(4) $3\pi - 30 \log_e 2 + 20 \log_e 5$

Q6 - 2024 (05 Apr Shift 2)

Let $\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$, $m, n > 0$. If $\int_0^1 (1-x^{10})^{20} dx = a \times \beta(b, c)$, then $100(a + b + c)$ equals _____

(1) 1021

(2) 2120

(3) 2012

(4) 1120

Q7 - 2024 (05 Apr Shift 2)

If $f(t) = \int_0^\pi \frac{2x \, dx}{1 - \cos^2 t \sin^2 x}$, $0 < t < \pi$, then the value of $\int_0^{\frac{\pi}{2}} \frac{\pi^2 dt}{f(t)}$ equals _____

Q8 - 2024 (06 Apr Shift 1)

$\int_0^{\pi/4} \frac{\cos^2 x \sin^2 x}{(\cos^3 x + \sin^3 x)^2} dx$ is equal to

(1) $1/6$

(2) $1/3$

(3) $1/12$

(4) $1/9$

Q9 - 2024 (06 Apr Shift 1)

Let $r_k = \frac{\int_0^1 (1-x^7)^k dx}{\int_0^1 (1-x^7)^{k+1} dx}$, $k \in \mathbb{N}$. Then the value of $\sum_{k=1}^{10} \frac{1}{7(r_k - 1)}$ is equal to _____

Q10 - 2024 (06 Apr Shift 2)

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Let $[t]$ denote the largest integer less than or equal to t . If

$\int_0^3 \left([x^2] + \left[\frac{x^2}{2} \right] \right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}$, where $a, b, c \in \mathbf{Z}$, then $a + b + c$ is equal to _____

Q11 - 2024 (08 Apr Shift 1)

The value of $k \in \mathbf{N}$ for which the integral $I_n = \int_0^1 (1 - x^k)^n dx, n \in \mathbf{N}$, satisfies $147I_{20} = 148I_{21}$ is

- (1) 14
- (2) 8
- (3) 10
- (4) 7

Q12 - 2024 (08 Apr Shift 2)

Let $\int_{\alpha}^{\log_e 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$. Then e^{α} and $e^{-\alpha}$ are the roots of the equation :

- (1) $x^2 + 2x - 8 = 0$
- (2) $x^2 - 2x - 8 = 0$
- (3) $2x^2 - 5x + 2 = 0$
- (4) $2x^2 - 5x - 2 = 0$

Q13 - 2024 (09 Apr Shift 2)

Let $\int_0^x \sqrt{1 - (y'(t))^2} dt = \int_0^x y(t) dt, 0 \leq x \leq 3, y \geq 0, y(0) = 0$. Then at $x = 2, y'' + y + 1$ is equal to

- (1) 1
- (2) 2
- (3) $\sqrt{2}$
- (4) $1/2$

Q14 - 2024 (09 Apr Shift 2)

The value of the integral $\int_{-1}^2 \log_e (x + \sqrt{x^2 + 1}) dx$ is

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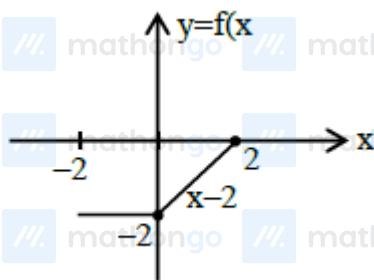
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Answer Key

Solutions

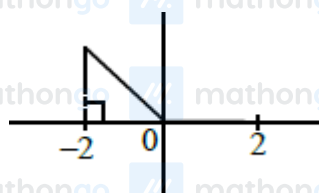
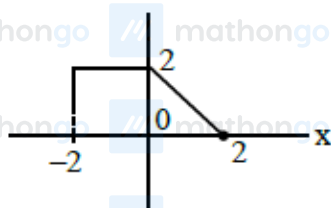
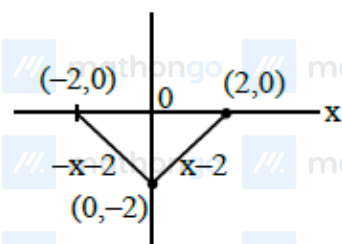
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Q1



$f(|x|) \rightarrow$

$|f(x)|$



$$h(x) = \begin{cases} x-2+2-x=0, & 0 \leq x \leq 2 \\ -x-2+2=-x & -2 \leq x < 0 \end{cases}$$

$$\Rightarrow \int_0^2 h(x)dx = 0 \text{ and } \int_{-2}^0 h(x)dx = 2$$

Q2

$$\int_0^{\frac{\pi}{2}} \frac{\sin^2 x}{1 + \frac{1}{2} \sin 2x} dx = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{2 + \sin 2x} dx$$

$$\int \frac{1}{2 + \sin 2x} - \int \frac{\cos 2x}{2 + \sin 2x}$$

$$(I_1) - (I_2)$$

$$(I_1) = \int \frac{dx}{2 + \frac{2 \tan x}{1 + \tan^2 x}}$$

$$\int_0^{\frac{\pi}{4}} \frac{\sec^2 x dx}{2 \tan^2 x + 2 \tan x + 2}$$

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Solutions

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$$\tan x = t$$

$$\frac{1}{2} \int_0^1 \frac{dt}{\left(t + \frac{1}{2}\right)^2 + \frac{3}{4}} = \frac{\pi}{6\sqrt{3}}$$

$$I_2 = \int_0^{\pi/4} \frac{\cos 2x}{2 + \sin 2x} dx = \frac{1}{2} \left(\ln \frac{3}{2} \right)$$

$$I_1 - I_2 = \frac{1}{\sqrt{3}} \frac{\pi}{6} + \frac{1}{2} \ln \frac{2}{3}$$

$$\Rightarrow a = 2, b = 6$$

Ans. 8

Q3

Let $I = \int_{-1}^{+1} \frac{\cos \alpha x}{1+3^x} dx \dots (I)$

$$I = \int_{-1}^{+1} \frac{\cos \alpha x}{1+3^{-x}} dx$$

$$\left(u \sin \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right) \dots (II)$$

Add (1) and (II)

$$2I = \int_{-1}^{+1} \cos(\alpha x) dx = 2 \int_0^1 \cos(\alpha x) dx$$

$$I = \frac{\sin \alpha}{\alpha} = \frac{2}{\pi} \text{ (given)}$$

$$\therefore \alpha = \frac{\pi}{2}$$

Q4

$$\int_{-\pi}^{\pi} \frac{2y(1+\sin y)}{1+\cos^2 y} dy$$

$$= \int_{-\pi}^{\pi} \frac{2y}{1+\cos^2 y} dy + \int_{-\pi}^{\pi} \frac{2y \sin y}{1+\cos^2 y} dy$$

(Odd) (Even)

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Solutions

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$$= 0 + 2.2 \int_0^{\pi} y \left(\frac{\sin y}{1 + \cos^2 y} \right) dy$$

$$I = 4 \int_0^{\pi} \frac{y \sin y}{1 + \cos^2 y} dy$$

$$I = 4 \int_0^{\pi} \frac{(\pi - y) \sin y}{1 + \cos^2 y} dy$$

$$2I = 4 \int_0^{\pi} \frac{\pi \sin y}{1 + \cos^2 y} dy$$

$$I = 2\pi \int_0^{\pi} \frac{\sin y}{1 + \cos^2 y} dy$$

$$= 2\pi \left(-\tan^{-1}(\cos y) \right)_0^{\pi}$$

$$= -2\pi \left[\left(-\frac{\pi}{4} \right) - \left(\frac{\pi}{4} \right) \right]$$

$$= -2\pi \left[-\frac{2\pi}{4} \right] = \pi^2$$

Q5

$$I = \int_0^{\pi/4} \frac{136 \sin x}{3 \sin x + 5 \cos x} dx$$

$$136 \sin x = A(3 \sin x + 5 \cos x) + B(3 \cos x - 5 \sin x)$$

$$136 = 3A - 5B \dots (i)$$

$$0 = 5A + 3B \dots (ii)$$

$$3B = -5A \Rightarrow B = -\frac{5}{3}A$$

$$136 = 3A - 5 \left(-\frac{5}{3}A \right)$$

$$136 = 3A + \frac{25}{3}A$$

$$136 = \frac{34A}{3}$$

$$\Rightarrow A = \frac{136 \times 3}{34} = 12$$

$$B = -\frac{5}{3}(12) = -20$$

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Solutions

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$$\begin{aligned}
 I &= \int_0^{\pi/4} \frac{A(3 \sin x + 5 \cos x)}{3 \sin x + 5 \cos x} + \int_0^{\pi/4} \frac{B(3 \cos x - 5 \sin x)}{3 \sin x + 5 \cos x} \\
 &= A(x)_0^{\pi/4} + B[\ln(3 \sin x + 5 \cos x)]_0^{\pi/4} \\
 &= 12 \left(\frac{\pi}{4} \right) - 20 \ln \left(\frac{3}{\sqrt{2}} + \frac{5}{\sqrt{2}} \right) - \ln(0 + 5) \\
 &= 3\pi - 20 \ln 4\sqrt{2} + 20 \ln 5 \\
 &= 3\pi - 20 \times \frac{5}{2} \ln 2 + 20 \ln 5 \\
 &= 3\pi - 50 \ln 2 + 20 \ln 5
 \end{aligned}$$

Q6

$$I = \int_0^1 1 \cdot (1 - x^{10})^{20} dx$$

$$x^{10} = t$$

$$x = t^{1/10}$$

$$dx = \frac{1}{10} (t)^{-9/10} dt$$

$$I = \int_0^1 (1 - t)^{20} \frac{1}{10} (t)^{-9/10} dt$$

$$I = \frac{1}{10} \int_0^1 t^{-9/10} (1 - t)^{20} dt$$

$$a = \frac{1}{10} \quad b = \frac{1}{10} \quad c = 21$$

Q7

$$f(t) = \int_0^\pi \frac{2x}{1 - \cos^2 t \sin^2 x} dx \dots (1)$$

$$= 2 \int_0^\pi \frac{(\pi - x) dx}{1 - \cos^2 t \sin^2 x} \dots (2)$$

$$2f(t) = 2 \int_0^\pi \frac{\pi}{1 - \cos^2 t \sin^2 x} dx$$

$$f(t) = \int_0^\pi \frac{\pi}{1 - \cos^2 t \sin^2 x} dx$$

divide by $\cos^2 x$

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Solutions

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$$f(t) = \pi \int_0^{\pi} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t^2 x}$$

$$f(t) = 2\pi \int_0^{\pi/2} \frac{\sec^2 x dx}{\sec^2 x - \cos^2 t^2 \tan^2 x}$$

$$\tan x = z$$

$$\sec^2 x dx = dz$$

$$f(t) = 2\pi \int_0^{\infty} \frac{dz}{1 + \sin^2 t \cdot z^2}$$

$$= \frac{\pi^2}{\sin t}$$

$$\text{Then } \int_0^{\pi/2} \frac{\pi^2}{f(t)} dt$$

$$= \int_0^{\pi/2} \sin t dt$$

$$= 1$$

Q8

Divide Nr&Dr by $\cos x$

$$\int_0^{\pi/4} \frac{\tan^2 x \sec^2 x dx}{(1 + \tan^3 x)^2} dx$$

$$\text{Let } 1 + \tan^3 x = t$$

$$\tan^2 x \sec^2 x dx = \frac{dt}{3}$$

$$\frac{1}{3} \int_1^2 \frac{dt}{t^2} = \frac{1}{6}$$

Q9

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Solutions

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$$I_K = \int_0^1 1 \cdot (1 - x^7)^K dx$$

$$I_K = (1 - x^7)^K x \Big|_0^1 + 7K \int_0^1 (1 - x^7)^{K-1} x^6 \cdot x dx$$

$$I_K = -7K \int_0^1 (1 - x^7)^{K-1} ((1 - x^7) - 1) dx$$

$$I_K = -7KI_K + 7KI_{K-1}$$

$$\Rightarrow \frac{I_K}{I_{K+1}} = \frac{7K+8}{7K+7}$$

$$r_K = \frac{7K+8}{7K+7}$$

$$r_K - 1 = \frac{1}{7(K+1)}$$

$$\Rightarrow 7(r_K - 1) = \frac{1}{K+1}$$

$$\sum_{K=1}^{10} (K+1) = 11(6) - 1 = 65$$

Q10

$$\int_0^3 [x^2] dx + \int_0^3 \left[\frac{x^2}{2} \right] dx$$

$$= \int_0^1 0 dx + \int_1^{12} 1 dx + \int_{\sqrt{2}}^{\sqrt{3}} 2 dx$$

$$+ \int_{\sqrt{3}}^2 3 dx + \int_2^{\sqrt{5}} 4 dx + \int_{\sqrt{5}}^{\sqrt{6}} 5 dx$$

$$+ \int_{\sqrt{6}}^{\sqrt{7}} 6 dx + \int_{\sqrt{7}}^{\sqrt{8}} 7 dx + \int_{\sqrt{8}}^3 8 dx$$

$$+ \int_0^{\sqrt{2}} 0 dx + \int_{\sqrt{2}}^2 1 dx$$

$$+ \int_2^{\sqrt{6}} 2 dx + \int_{\sqrt{6}}^{\sqrt{8}} 3 dx + \int_{\sqrt{8}}^3 4 dx = 31 - 6\sqrt{2} - \sqrt{3} - \sqrt{5}$$

$$- 2\sqrt{6} - \sqrt{7}$$

$$a = 31 \quad b = -6 \quad c = -2$$

$$a + b + c = 31 - 6 - 2 = 23$$

Q11

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Solutions

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$$I_n = \int_0^1 (1-x^k)^n \cdot 1 dx$$

$$I_n = (1-x^k)^n \cdot x - nk \int_0^1 (1-x^k)^{n-1} \cdot x^{k-1} \cdot dx$$

$$I_n = nk \int_0^1 \left[(1-x^k)^n - (1-x^k)^{n-1} \right] dx$$

$$I_n = nk I_n - nk I_{n-1}$$

$$\frac{I_n}{I_{n-1}} = \frac{nk}{nk+1}$$

$$\frac{I_{21}}{I_{20}} = \frac{21k}{1+21k}$$

$$= \frac{147}{148} \Rightarrow k = 7$$

Q12

$$\int_{\alpha}^{\log 4} \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$$

$$\text{Let } e^x - 1 = t^2$$

$$e^x dx = 2t dt$$

$$= \int \frac{2t dt}{t^2 + 1}$$

$$= 2 \tan^{-1} t$$

$$= 2 \tan^{-1} \left(\sqrt{e^x - 1} \right) \Big|_{\alpha}^{\log 4}$$

$$= 2 \left[\tan^{-1} \sqrt{3} - \tan^{-1} \sqrt{e^{\alpha} - 1} \right] = \frac{\pi}{6}$$

$$= \frac{\pi}{3} - \tan^{-1} \sqrt{e^{\alpha} - 1} = \frac{\pi}{12}$$

$$\Rightarrow \tan^{-1} \sqrt{e^{\alpha} - 1} = \frac{\pi}{4}$$

$$e^{\alpha} = 2 \quad e^{-\alpha} = \frac{1}{2}$$

$$x^2 - \left(2 + \frac{1}{2} \right) x + 1 = 0$$

$$2x^2 - 5x + 2 = 0$$

Q13

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Solutions

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$$\sqrt{1 - (y'(x))^2} = y(x)$$

$$1 - \left(\frac{dy}{dx}\right)^2 = y^2$$

$$\left(\frac{dy}{dx}\right)^2 = 1 - y^2$$

$$\frac{dy}{\sqrt{1 - y^2}} = dx \text{ OR } \frac{dy}{\sqrt{1 - y^2}} = -dx$$

$$\Rightarrow \sin^{-1} y = x + c, \sin^{-1} y = -x + c$$

$$x = 0, y = 0 \Rightarrow c = 0$$

$$\sin^{-1} y = x, \text{ as } y \geq 0$$

$$\sin x = y$$

$$\Rightarrow \frac{dy}{dx} = \cos x$$

$$\frac{d^2y}{dx^2} = -\sin x$$

$$\Rightarrow -\sin x + \sin x + 1 = 1$$

Q14

$$I = \int_{-1}^2 \log_e(x + \sqrt{x^2 + 1}) dx$$

$$= x \log_e(x + \sqrt{x^2 + 1}) - \int_{-1}^2 \left(\frac{1 + \frac{x}{\sqrt{x^2 + 1}}}{x + \sqrt{x^2 + 1}} \right) dx$$

$$= x \log_e(x + \sqrt{x^2 + 1}) - \int_{-1}^2 \frac{x}{\sqrt{x^2 + 1}} dx$$

$$= x \log_e(x + \sqrt{x^2 + 1}) - \sqrt{x^2 + 1} \Big|_{-1}^2$$

$$= (2 \log_e(2 + \sqrt{5}) - \sqrt{5})$$

$$- (-\log_e(-1 + \sqrt{2}) - \sqrt{2})$$

$$= \log_e(2 + \sqrt{5})^2 - \sqrt{5} + \log_e(\sqrt{2} - 1) + \sqrt{2}$$

$$= \log_e(2 + \sqrt{5})^2 - \sqrt{5} + \log_e(\sqrt{2} - 1) + \sqrt{2}$$

$$= \sqrt{2} - \sqrt{5} + \log_e\left(\frac{(2 + \sqrt{5})^2}{\sqrt{2} + 1}\right)$$

$$= \sqrt{2} - \sqrt{5} + \log_e\left(\frac{9 + 4\sqrt{5}}{\sqrt{2} + 1}\right)$$

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